

Task:1

```
21 </html> -->
22 <!DOCTYPE html>
23 <html lang="en">
24 <head>
25   <meta charset="UTF-8">
26   <meta name="viewport" content="width=device-width, initial-scale=1.0">
27   <title>Document</title>
28 </head>
29 <body>
30   <script>
31     const Student={
32       name: "Riya",
33       rollno: 101,
34       marks: {Math: 85, Science: 90, English: 78, History: 88, Art: 92 },
35     };
36     const {Math: math, Science: science, English: english, History: history, Art: art} = Student.marks;
37     const total=math + science + english + history + art;
38     const percentage= (total/500)*100;
39
40     const report= `Report Card for ${Student.name} (Roll No: ${Student.rollno})
41     -----
42     Subjects: Math: ${math}, Science: ${science}, English: ${english}, History: ${history}, Art: ${art}
43     Total Marks: ${total}
44     Percentage: ${percentage}%`;
45     console.log(report);
46   </script>
47 </body>
48 </html>
49 <!-- <!DOCTYPE html>
```

Report Card for Riya (Roll No: 101)

Subjects: Math: 85, Science: 90, English: 78, History: 88, Art: 92

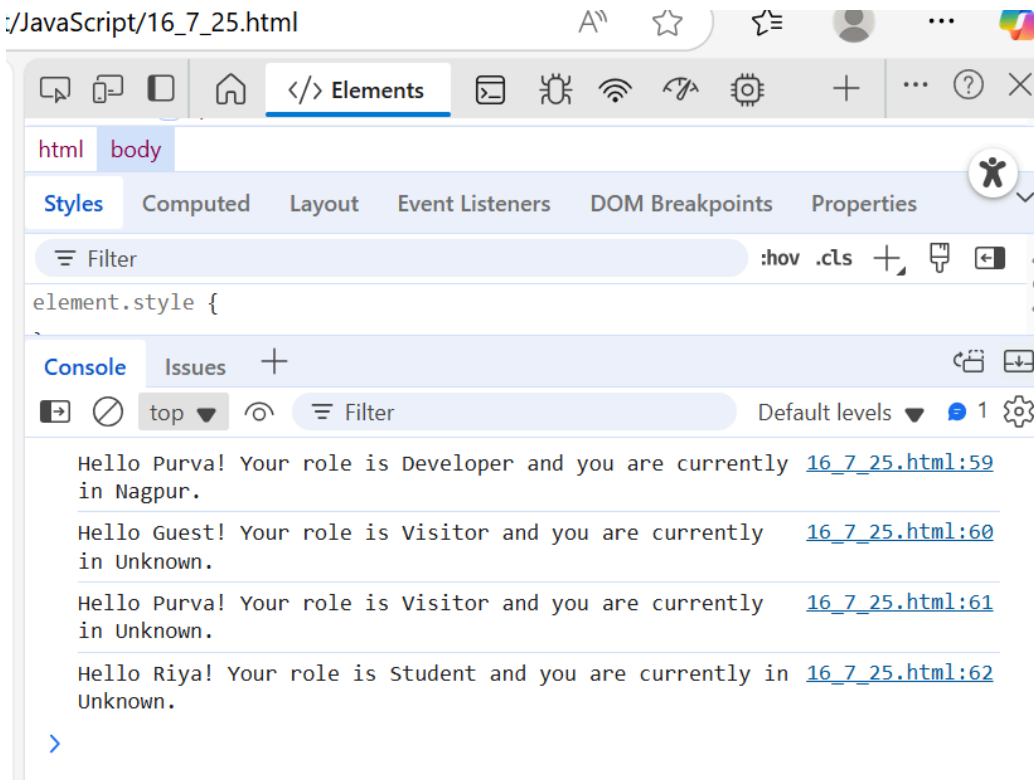
Total Marks: 433

Percentage: 86.6%

Task:2

```
JavaScript > <> 16_7_25.html > ...
49 | <!DOCTYPE html>
50 | <html lang="en">
51 | <head>
52 |   <title>Greeting Generator</title>
53 | </head>
54 | <body>
55 |   <script>
56 |     const generateGreeting = (name = "Guest", role = "Visitor", location = "Unknown") => {
57 |       return `Hello ${name}! Your role is ${role} and you are currently in ${location}.`;
58 |     };
59 |     console.log(generateGreeting("Purva", "Developer", "Nagpur"));
60 |     console.log(generateGreeting());
61 |     console.log(generateGreeting("Purva"));
62 |     console.log(generateGreeting("Riya", "Student"));
63 |   </script>
64 | </body>
65 | </html>
```

:/JavaScript/16_7_25.html



Task:3

```
Welcome  Lab1.html  Lab2.html  JS lab2.js  SITPage.html  16_7_25.html  Info_Page.html
JavaScript > 16_7_25.html > html > body > script > analyzeData
85 <!DOCTYPE html>
86 <html lang="en">
87 <body>
88 <script>
89 const analyzeData = (input) => {
90   let type = "";
91   let mutable = "";
92   if (input === null) {
93     type = "null";
94     mutable = "immutable";
95   } else if (Array.isArray(input)) {
96     type = "array";
97     mutable = "mutable";
98   } else if (input instanceof Function) {
99     type = "function";
100    mutable = "mutable";
101   } else if (input instanceof String || typeof input === "string") {
102     type = "string";
103     mutable = "immutable";
104   } else if (input instanceof Number || Object.prototype.toString.call(input) === "[object Number]") {
105     type = "number";
106     mutable = "immutable";
107   } else if (input instanceof Boolean || Object.prototype.toString.call(input) === "[object Boolean]") {
108     type = "boolean";
109     mutable = "immutable";
110   } else if (input instanceof Symbol) {
111     type = "symbol";
112     mutable = "immutable";
113   } else if (input instanceof BigInt) {
114     type = "bigint";
```

Welcome Lab1.html lab2.html JS lab2.js SITPage.html 1

JavaScript > 16_7_25.html > html > body > script > analyzeData

```
86 <html lang="en">
87 <body>
88 <script>
89 const analyzeData = (input) => {
111     type = "symbol";
112     mutable = "immutable";
113 } else if (input instanceof BigInt) {
114     type = "bigint";
115     mutable = "immutable";
116 } else if (input === undefined) {
117     type = "undefined";
118     mutable = "immutable";
119 } else if (input instanceof Object) {
120     type = "object";
121     mutable = "mutable";
122 } else {
123     type = "unknown";
124     mutable = "unknown";
125 }
126 console.log(`Type: ${type}`);
127 console.log(`Mutability: ${mutable}`);
128 console.log(`Summary: This value is a(n) ${type} and it is ${mutable}.`);
129 };
130 analyzeData("hello");
131 analyzeData(123);
132 analyzeData(true);
133 analyzeData(null);
134 analyzeData(undefined);
135 analyzeData([1, 2, 3]);
```

Welcome

<> Lab1.html

<> lab2.html

JS lab2.js

JavaScript > <> 16_7_25.html > html > body > script > analyzeData

86 <html lang="en">

87 <body>

88 <script>

130 analyzeData("hello");

131 analyzeData(123);

132 analyzeData(true);

133 analyzeData(null);

134 analyzeData(undefined);

135 analyzeData([1, 2, 3]);

136 analyzeData({ name: "Riya" });

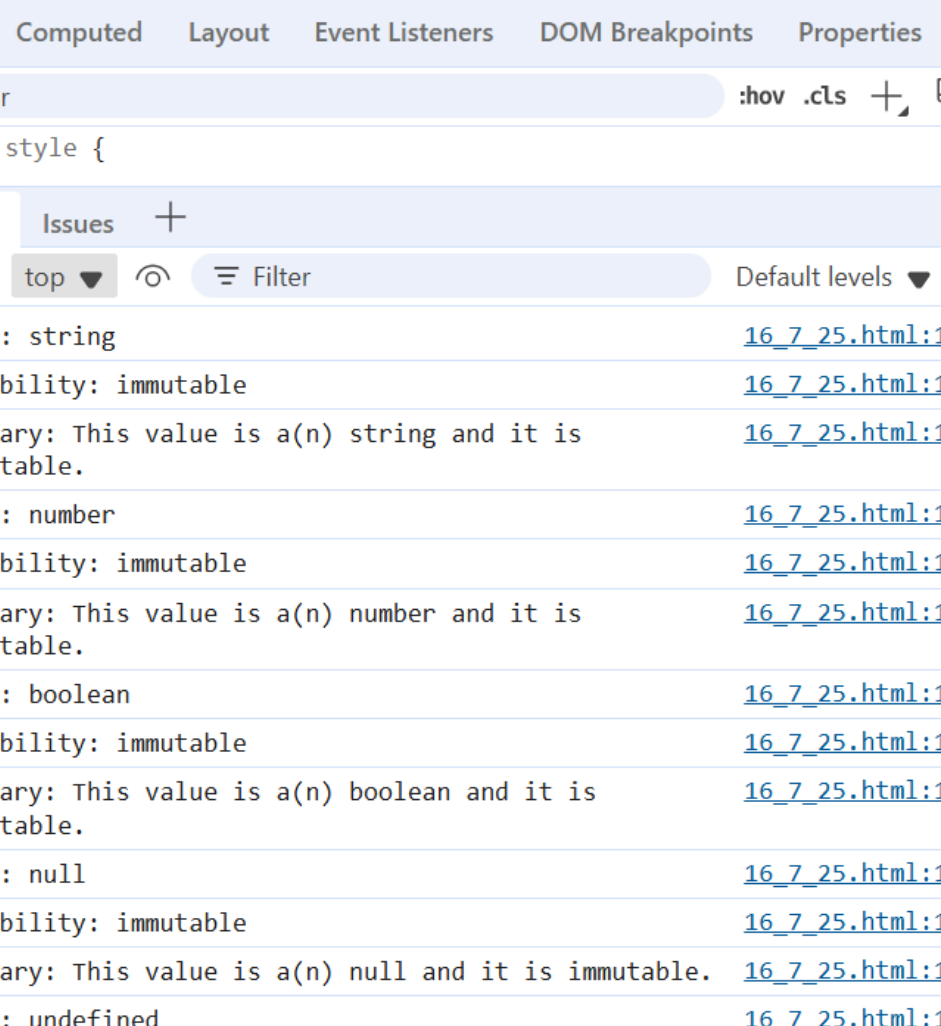
137 analyzeData(() => {});

138 </script>

139 </body>

140 </html>

141



The screenshot shows the Chrome DevTools console with the 'element.style' object selected. The console displays the object's prototype chain and its properties. The properties shown are:

- Type: string
- Mutability: immutable
- Summary: This value is a(n) string and it is immutable.
- Type: number
- Mutability: immutable
- Summary: This value is a(n) number and it is immutable.
- Type: boolean
- Mutability: immutable
- Summary: This value is a(n) boolean and it is immutable.
- Type: null
- Mutability: immutable
- Summary: This value is a(n) null and it is immutable.
- Type: undefined
- Mutability: immutable
- Summary: This value is a(n) undefined and it is immutable.

The console also shows the 'element.style' object's prototype chain, including 'ElementStyle' and 'Object'.

html body

Styles Computed Layout Event Listeners DOM Breakpoints Properties

Filter :hov .cls +

element.style {

Console Issues +

top Filter Default levels 2

Type: null	16_7_25.html:126
Mutability: immutable	16_7_25.html:127
Summary: This value is a(n) null and it is immutable.	16_7_25.html:128
Type: undefined	16_7_25.html:126
Mutability: immutable	16_7_25.html:127
Summary: This value is a(n) undefined and it is immutable.	16_7_25.html:128
Type: array	16_7_25.html:126
Mutability: mutable	16_7_25.html:127
Summary: This value is a(n) array and it is mutable.	16_7_25.html:128
Type: object	16_7_25.html:126
Mutability: mutable	16_7_25.html:127
Summary: This value is a(n) object and it is mutable.	16_7_25.html:128
Type: function	16_7_25.html:126
Mutability: mutable	16_7_25.html:127
Summary: This value is a(n) function and it is mutable.	16_7_25.html:128

>

Task:4

```
Welcome  Lab1.html  lab2.html  JS lab2.js  SITPage.html  16_7_25.html
JavaScript > 16_7_25.html > html > body > script > message
66  <!DOCTYPE html>
67  <html lang="en">
68  <body>
69      <script>
70          const colors1 = ["red", "green"];
71          const colors2 = ["blue", "yellow"];
72          const palette = [...colors1, ...colors2];
73          const [first, ...rest] = palette;
74          const [last] = rest.slice(-1);
75          const middle = rest.slice(0, -1);
76          const message = `Main colors: ${capitalize(first)} and ${capitalize(last)}.
77          Others in the palette: ${middle.map(capitalize).join(", ")}`;
78
79          console.log(message);
80          function capitalize(color) {
81              return color.charAt(0).toUpperCase() + color.slice(1);
82          }
83      </script>
84  </body>
85  </html>
```